

Dhruba Jyoti Seal

Artificial Intelligence Engineer

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GitHub - <https://github.com/dhrubaseal>

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Summary

Artificial Intelligence Engineer with a strong background in Data Science, Statistics, Machine Learning, Deep Learning, Computer Vision & Natural Language Processing. Currently working on Cyber Security domain.

Skills & Technologies

AI / ML & LLMs

Frameworks & Pipelines: RAG Systems, LangGraph, LangChain, Semantic Chunking

LLM Integration: Claude / Anthropic, GPT-4 / OpenAI, Ollama models

Multi-Agent & Orchestration: Multi-Agent Orchestration, AI Agents (A2A, HTTP), MCP Servers (FastMCP, stdio)

Reasoning: Chain-of-Thought, Tree-of-Thought

Vector Databases: ChromaDB, Qdrant, FAISS

Embeddings: FastEmbed, OpenAI text-embedding-3-large

Machine Learning & Deep Learning

Frameworks: TensorFlow, PyTorch, Scikit-learn, MLPClassifier, RandomForestRegressor

Computer Vision: YOLO, MediaPipe, OpenCV, EasyOCR

Audio & Translation: Speech Recognition, gTTS, M2M100

Annotation: Labelimg, CVAT, Labelme

Backend & Infrastructure

Frameworks & APIs: FastAPI, Flask, SQLAlchemy, REST APIs, asyncio, Multithreading, Queue Management, Real-time Processing

Authentication: JWT, OTP Systems, Session Management

Databases & Storage: PostgreSQL, MongoDB, Redis, MinIO

DevOps: Docker, Docker Compose, Kubernetes

Email: SMTP, Privacy-focused Distribution Systems

Frontend & Data

UI Frameworks: React, Streamlit, Tkinter

Data Processing: NumPy, Pandas, BeautifulSoup

Visualization: Power BI, Matplotlib, Seaborn

Media: ffmpeg

Security & Cloud

AWS: IAM, S3, EC2, VPC, KMS, CloudTrail, GuardDuty, Security Hub, boto3

Threat Intelligence: VirusTotal, GreyNoise, Shodan, HavelBeenPwned, CVE Analysis, Automated Threat Scoring, Real-time Enrichment

Tools & Platforms

Languages: Python

Version Control: Git, GitHub

OS: Linux

Financial: Thomson Reuters Eikon, Refinitiv Workspace

System: Multithreading, Queue Management, Real-time Processing

Projects

Machine Learning Engineer

Starinco | Sept 2023 – Dec 2023

- Proficient in annotation tools such as Labelimg, Labelme, and CVAT for image and video datasets.
- Manage datasets, break them down into train/validation/test sets, and translate between different markup formats.
- Experienced in running computer vision programs for images and videos.
- Competent in configuring and using workflow models with a strong understanding of configuration files.
- Solid foundation in Linux commands for tasks such as installation of CUDA drivers and system services.
- Proficient in using ffmpeg for slicing, translating formats, handling RTSP streams, and managing files and directories.
- Skilled in Python scripting for diverse tasks including collecting statistics on classes, dataset analysis, drawing detections, and interacting with services through JSON.
- Familiarity with dataset management, augmentations, and training pipelines including YOLO.
- Proficient in version control systems including Git, with experience in merging and collaborating on code repositories.
- Hands-on experience in installing, configuring, and managing RTMIP (Real Time Media Identification Platform).

- Expertise in implementing analytics logic, adding cameras, running neural services, and using the platform's dashboard and graphics.
- Strong foundation in system administration tasks including mounting hard disks, managing folders with tools like rsync, scp, sftp, and understanding VPNs.

Machine Learning Engineer

Genie AI | Jun 2024 – Oct 2024

- Design, develop, and implement machine learning architectures for AI assistant and other AI applications.
- Develop NLP chatbot based on proprietary datasets.
- Train, test, and optimize NLP models to improve understanding and generation of natural language.
- Collaborate with data scientists and AI researchers to develop innovative algorithms and techniques for AI assistant capabilities.
- Work closely with software engineers to integrate machine learning models into production systems.
- Ensure the scalability, reliability, and performance of machine learning and NLP components.
- Develop and maintain documentation for machine learning architectures and NLP models, including technical specifications, design patterns, and best practices.
- Stay current with industry trends and emerging technologies, applying new knowledge to enhance and evolve AI assistant solutions.

Professional Experience

Artificial Intelligence Engineer

Ampcus Cyber | May 2025 – Present

GRC Compliance Platform — RAG & LLM Integration

- **Architected a production RAG pipeline** for a GRC compliance platform enabling natural language querying over 7+ frameworks (PCI DSS, ISO 27001, SOC 2, NIST CSF, HIPAA, GDPR, PCI PIN) with citation-grounded answers using Anthropic Claude and OpenAI GPT models, achieving 95%+ query accuracy.
- **Designed a multi-provider LLM failover architecture** with primary (Anthropic) and secondary (OpenAI) providers, implementing automatic retry logic on rate limits, auth failures, and 5xx errors to ensure 99%+ AI service availability.
- **Built a dual vector store abstraction layer** supporting ChromaDB and Qdrant with configurable local (FastEmbed/mxbai-embed-large-v1) and cloud (OpenAI text-embedding-3-large) embeddings, enabling flexible deployment from local dev to cloud-scale production.
- **Developed an end-to-end document processing pipeline** supporting PDF, DOCX, XLSX, TXT, and image formats with OCR via OpenAI Vision API, semantic chunking via LangChain RecursiveCharacterTextSplitter, and automatic chunk classification (requirement, guidance, example, definition) for optimized retrieval.
- **Designed an AI response caching layer** with configurable TTL (30-day default) and automatic cleanup, reducing redundant LLM calls and lowering operational costs for frequently queried compliance topics.

AI Agents & Orchestration

- **Implemented a stateful LangGraph compliance assessment workflow** with a multi-node state machine (initialize → analyze → CoT/ToT reasoning → gap analysis → human review → report generation) and optional PostgreSQL checkpointing for long-running assessments.
- **Engineered an intelligent reasoning engine** with heuristic-based routing between Chain-of-Thought (CoT) and Tree-of-Thought (ToT) strategies, dynamically selecting reasoning depth based on query complexity, business impact, and multi-framework analysis requirements.
- **Built a real-time LLM cost optimization system** with in-memory token tracking and budget guardrails that automatically downgrades ToT to CoT when approaching monthly spend limits, balancing cost efficiency with output quality.
- **Designed asynchronous AI evidence indexing pipeline** triggered on upload events using background async tasks to perform vectorization without blocking user-facing API responses.

MCP Server & Platform Engineering

- **Built a production-grade MCP server** exposing 43 read-only AWS compliance tools via the Model Context Protocol (FastMCP SDK, stdio transport), enabling AI assistants (Cursor, Claude) to query real-time AWS infrastructure state across IAM, S3, EC2, VPC, KMS, CloudTrail, GuardDuty, Security Hub, and 35+ services.
- **Engineered a secure AWS integration layer** with a whitelisted boto3 client factory enforcing read-only access, comprehensive IAM policy definitions, and a sample data mode (USE_SAMPLE_DATA=true) for safe E2E testing without live AWS credentials.
- **Architected a DDD modular monolith backend (FastAPI/Python)** with 10+ domain modules communicating via Module APIs and DTOs, integrated with a RAG microservice, MongoDB, Redis, and MinIO; containerized the full stack with Docker Compose.
- **Developed an AI-powered evidence review system** that automatically indexes uploaded evidence, runs compliance analysis against framework-specific controls, and returns structured verdicts (meets/partial/does_not_meet/insufficient) with confidence scores, gap analysis, and remediation recommendations.

- **Implemented a multi-turn RAG chat interface** for evidence-based compliance conversations, supporting per-organization evidence collections, context-aware follow-up questions, and source citations from compliance standards and uploaded evidence documents.
- **Built a modular cybersecurity intelligence platform** aggregating real-time threat data via VirusTotal, GreyNoise, Shodan, and HaveIBeenPwned APIs with LLM-powered summarization, automated CVE scoring, and a privacy-focused BCC email distribution system for daily security digests.

Machine Learning Engineer

CornerstoneX | Oct 2024 – Nov 2024

- Developed a multilingual voice assistant supporting 19+ languages using Python, integrating speech recognition, OpenAI's GPT-3.5, and M2M100 translation models, enabling real-time language detection and seamless cross-lingual communication.
- Engineered a real-time sign language detection system using YOLO object detection and MediaPipe, incorporating machine learning (MLPClassifier) for gesture recognition with GPT-4 integration for natural language responses.
- Built a smart media processing application integrating computer vision (OpenCV), OCR (EasyOCR), and YOLO object detection with real-time audio feedback using text-to-speech synthesis.
- Architected and implemented an end-to-end virtual dietitian platform using Python, leveraging RandomForestRegressor for weight progression modelling, FAISS vector indexing for real-time knowledge retrieval, OpenAI GPT-3.5 for NLP processing, PyTorch-based neural networks for nutrient optimization, and integrated custom data classes for type-safe user profiling and food item management through a microservices-style OOP architecture.
- Implemented multithreaded processing architecture using Python's threading and queue systems to ensure smooth real-time performance in video processing applications.
- Created responsive GUI applications using Tkinter and Streamlit, featuring real-time visualization, custom controls, and efficient state management.

System Engineer

Tata Consultancy Services | May 2022 – Oct 2022

- Used data science techniques to detect and resolve server issues, reducing system downtime.
- Optimized alarm settings using data analytics, resulting in a reduction in false alarms.
- Implemented predictive analysis for server alarms, resulting in faster response times and improved system performance.
- Developed an asset tracking system using data science tools and created a Power BI Dashboard for the infrastructure teams, managers, and leadership teams for better visibility of the Data Centers Inventory.
- Collaborated with infrastructure teams, managers, and leadership teams to develop and implement data governance policies and standard operating procedures.

System Engineer

Refinitiv, an LSEG business | Oct 2018 – May 2022

- Monitored various systems and tools to ensure they are functioning properly.
- Responded to alerts and issues by creating tickets and following defined processes.
- Troubleshoot Incident Management and resolved incidents based on established standard operating procedures (SOPs).
- Performed regular health checks of systems to evaluate their performance and capacity.
- Documented defects and maintained a knowledge database for known issues.
- Kept track of shift activities and facilitated handover between teams in different locations.
- Coordinated with other Site Reliability Engineers (SREs), ISO, and DB teams for knowledge transfer and other tasks.

System Security Administrator

State Street Global Advisors | Mar 2018 - Sep 2018

- Managed user accounts and access on a Microsoft Active Directory server-based environment, including user creation and termination.
- Set up and administered user and group mailboxes on Microsoft Exchange for many users.
- Configured secure shared folders with appropriate access controls and troubleshoot issues as needed.
- Managed access to resources through Group Policy Objects (GPOs) and role-based access controls for many role-specific groups.
- Created, managed, and synchronized batch and service accounts with on-call support.
- Managed authorized access to user, group, and system resources on various databases and platforms including Oracle, Unix, Sybase, IBM RACF, and Citrix Xen App.
- Managed privileged access to critical financial applications.
- Onboarded and configured over 45 applications to a centralized tool and managed ongoing maintenance and access.

Education

Bachelor of Technology

Electronics and Communications Engineering

West Bengal University of Technology | Jul 2012 - Aug 2016